ESC201: Mechanics of Materials

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Homework 6

Instructors: Professor Wootton

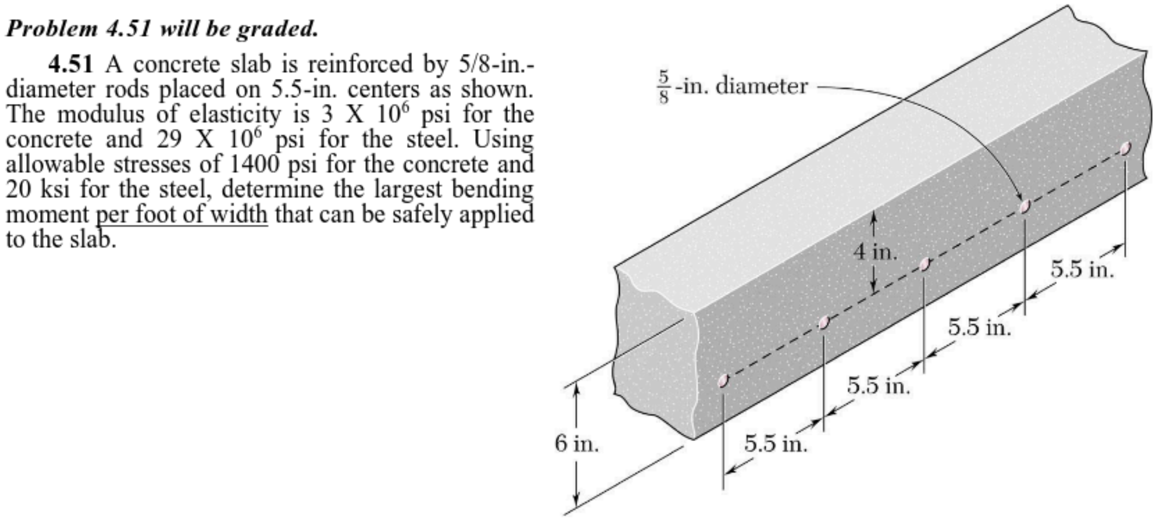
Time Spent: 4 Hours

March 30, 2026

Note that although all numbers present in this homework assignment are displayed with 2-4 significant digits, no rounding occurs until the final answer.

I apologize for the messy handwriting in my previous assignment, I hope that this medium is easier for you to parse through.

1. Problem 4.51



Problem 4.51 will be graded.

4.51 A concrete slab is reinforced by 5/8-in.-diameter rods placed on 5.5-in. centers as shown. The modulus of elasticity is 3 X 10^6 psi for the concrete and 29 X 10^6 psi for the steel. Using allowable stresses of 1400 psi for the concrete and 20 ksi for the steel, determine the largest bending moment per foot of width that can be safely applied to the slab.

Figure 1: *Problem Statement for Problem 4.51*

The following given figures are as follows:

$$d_s = \frac{5}{8} \text{ in}, \quad d = 4 \text{ in}, \quad l = 5.5 \text{ in}, \quad w = 6 \text{ in}, \quad E_s = 29 * 10^6 \text{ psi}, \quad E_c = 3 * 10^6 \text{ psi},$$

$$\sigma_{\text{all, c}} = 1.4 * 10^3 \text{ psi}, \quad \sigma_{\text{all, s}} = 20 \text{ ksi} = 2 * 10^4 \text{ psi}$$

Determine: $\max(M_{\text{ft}})$

Approach:

1. Express the system with two rectangles seperated by a factor of $(d - x)$.
2. Calculate the ratio $\left(n = \frac{E_s}{E_c}\right)$.
3. Locate the neutral axis x by solving the equation:

$$\frac{1}{2}lx^2 - nA_s(d - x) = 0$$

where $\left(A_s = \frac{\pi}{4}(d_s)^2\right)$.

4. Find the moment of inertia of the sys:

$$I = \frac{1}{3}lx^3 + nA_s(d - x)^2$$

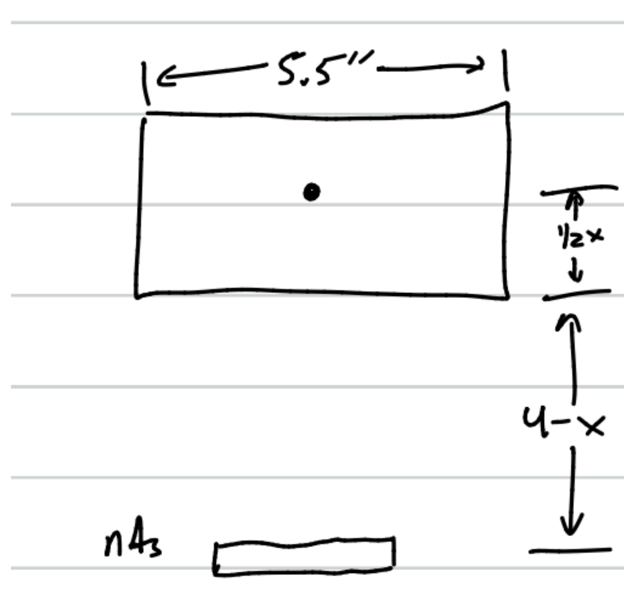


Figure 2: *Drawing of Equivalent System*

5. Rearrange the equation $\left(\sigma = \frac{nMy}{I}\right)$ into an equation for M to determine M_s and M_c.
6. Let $\max(M) = \min(\{M_s, M_c\})$.
7. Since M is the largest bending moment per width l, calculate the largest bending moment per foot of width M_{ft} with $\left(M_{\text{ft}} = \left(\frac{12}{l}\right)M\right)$.

Define n and A_s as:

$$n := \frac{E_s}{E_c}, \quad A_s := \frac{\pi}{4}(d_s)^2 \tag{1}$$

Then, using the appropiate given figures, we can determine x using the quadratic formula (note that this computation was done using the sympy Python library, see the Appendix Code in Section 4):

$$\left(\frac{1}{2}lx^2 - nA_s(d - x) = 0\right) \Rightarrow ((x = -2.685) \lor (x = 1.607)) \tag{2}$$

Note that x cannot be a negative value (as it would have no physical meaning in the context of this problem), so let $x = 1.607$ in.

Then, the moment of inertia of the system is:

$$I = \frac{1}{3}lx^3 + nA_s(d - x)^2 \tag{3}$$

Note that:

$$\left(\sigma = \frac{nMy}{I}\right) \Rightarrow \left(M = \frac{I\sigma}{ny}\right) \tag{4}$$

Therefore, by using Equation 4 relative to concrete $((n = 1) \land (y = x) \land (\sigma = \sigma_{\text{all, c}}))$:

$$M_c = \frac{I\sigma}{ny} = \frac{\left(\frac{1}{3}lx^3 + nA_s(d - x)^2\right)(\sigma_{\text{all, c}})}{x} = 2.143 * 10^4 \text{ lb} \cdot \text{in} \tag{5}$$

Additionally, by using Equation 4 relative to steel $((y = 4 - x) \land (\sigma = \sigma_{\text{all, s}}))$:

$$M_s = \frac{I\sigma}{ny} = \frac{\left(\frac{1}{3}lx^3 + nA_s(d - x)^2\right)(\sigma_{\text{all, s}})}{n(4 - x)} = 2.126 * 10^4 \text{ lb} \cdot \text{in} \tag{6}$$

Thus, the largest bending moment M per width l is $\left(M = 2.126 * 10^4 \text{ lb} \cdot \text{in}\right)$. Therefore, the largest bending moment M per foot of width is:

$$\boxed{M_{\text{ft}} = \left(\frac{12\text{ in}}{1\text{ ft}}\right)\left(\frac{M}{l}\right) = *\left(4.638 * 10^3\right)\frac{\text{lb} \cdot \text{in}}{\text{ft}} \qquad (7)}$$

<u>Discussion:</u> Note that $(M_s < M_c)$, meaning that the allowable bending moment for the reinforced concrete slab is governed by the steel stress limit. Also, since $(M_s \approx M_c)$, it implies that the design of this slab was efficient, so that the steel and concrete allowable stresses are reached at about the same moment.

2. Problem 4.65

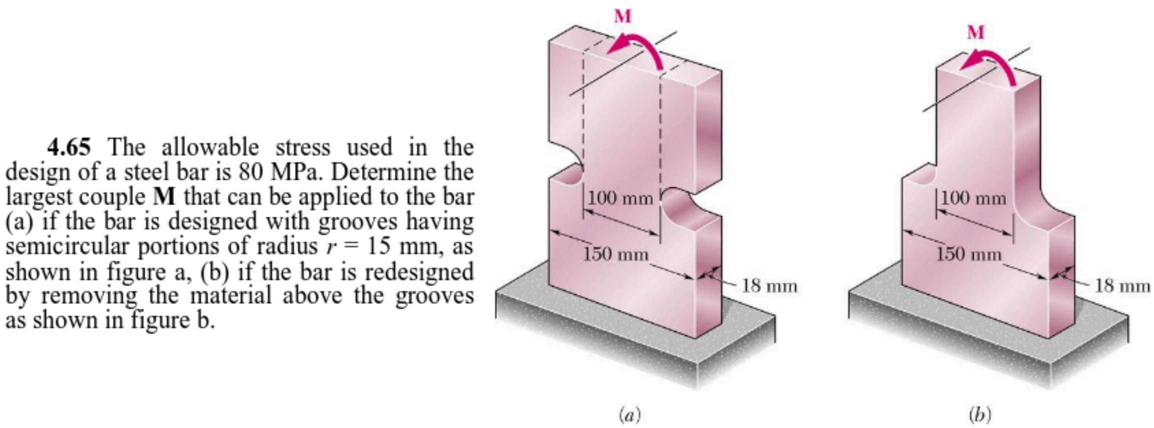


Figure 3: *Problem Statement for Problem 4.65*

The following given figures are as follows:

$$r = 15 \text{ mm} = 0.015 \text{ m}, \quad D = 150 \text{ mm} = 0.150 \text{ m}, \quad d = 100 \text{ mm} = 0.100 \text{ m},$$

$$w = 18 \text{ mm} = 0.018 \text{ m}, \quad \sigma_{\text{all}} = 80 \text{ MPa} = 8 * 10^7 \text{ Pa}$$

Determine:

a) $\max(M_a)$

b) $\max(M_b)$

<u>Approach:</u>

1. Calculate $\left(\frac{D}{d}\right)$ and $\left(\frac{r}{D}\right)$ ratios.
2. Using Fig 4.31 and Fig 4.32 from the textbook, determine the stress concentrations for both bar configurations K_a and K_b.
3. To find $\max(\sigma)$ calculate the minimum moment of inertia of both bars with $\left(I = \frac{1}{12}wd^3\right)$.
4. Rearrange $\left(\sigma = \frac{KMd}{2I}\right)$ into an equation for M to find the maximum moments for both bar configurations.

First, consider:

$$\frac{D}{d} = 1.5, \quad \frac{r}{d} = 0.15 \tag{8}$$

Then, by Fig 4.31 and Fig 4.32, respectively:

$$K_b = 1.58, \quad K_a = 1.92 \tag{9}$$

Also, to maximize the couple M, minimize the moment of inertia of the relevant cross section of the bar (which is the same cross section for both bar configurations), thus let:

$$I = \frac{1}{12}wd^3 \tag{10}$$

Note that:

$$\left(\sigma = \frac{KMd}{2I}\right) \Rightarrow \left(M = \frac{2\sigma I}{Kd}\right) \tag{11}$$

Then, for configuration (a):

$$\boxed{\max(M_a) = \frac{2\sigma_{\text{all}} I}{K_a d} = \left(1.250 * 10^3\right)\ \mathrm{N \cdot m}} \tag{12}$$

Lastly, for configuration (b):

$$\boxed{\max(M_b) = \frac{2\sigma_{\text{all}} I}{K_b d} = \left(1.519 * 10^3\right)\ \mathrm{N \cdot m}} \tag{13}$$

3. Problem 4.113

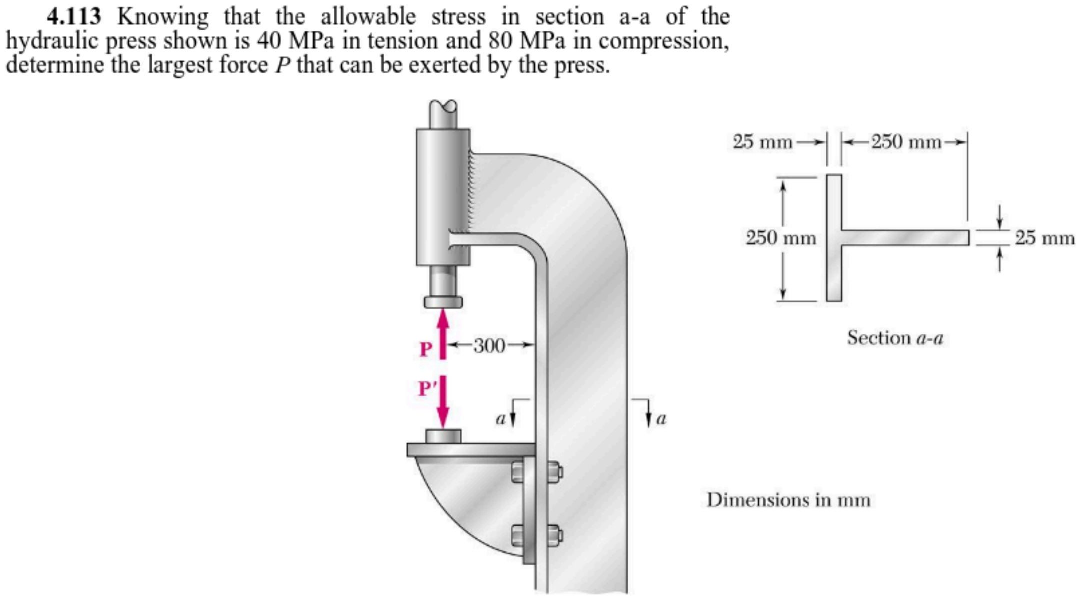


Figure 4: *Problem Statement for Problem 4.113*

The following given figures are as follows:

$$l_1 = 0.025 \text{ m}, \quad w_1 = 0.250 \text{ m}, \quad l_2 = 0.250 \text{ m}, \quad w_2 = 0.025 \text{ m}, \quad d = 0.300 \text{ m},$$

$$\sigma_{\text{all, c}} = -80 \text{ MPa} = -8 * 10^7 \text{ Pa}, \quad \sigma_{\text{all, t}} = 40 \text{ MPa} = 4 * 10^7 \text{ Pa}$$

Determine: $\max(P)$

<u>Approach:</u>

1. Find the centroid y_n of 1 and 2 by taking their midpoints (assuming that the object has uniform density $\left(\delta = (1)\frac{\text{kg}}{\text{m}^2}\right)$).
2. Take weighted averages of centroids of 1 and 2 with respect to their areas to find

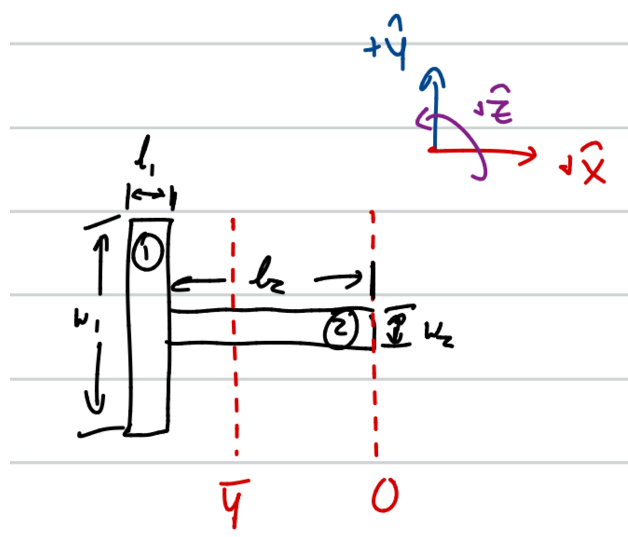


Figure 5: *Free Body Diagram of Cross Section of Press*

neutral axis $\overline{Y}$.

3. Find moment of inertia I_n of 1 and 2 via. $\left(I_n = \frac{1}{12}bh^3 + A_n d_n^2\right)$, then find the total moment of inertia I with $(I = I_1 + I_2)$.
4. To consider the eccentric loading of the hydraulic press, let $(M_e = -Pe)$, where $\left(e = l_1 + l_2 + \frac{d}{2} - \overline{Y}\right)$.
5. Considering the stress applied to 1 as the compressive stress and the stress applied to 2 as the tensile stress, use the total stress equation $\left(\sigma_n = \frac{P_n}{A} - \frac{My}{I}\right)$ to find an equation for P, and calculate the maximum force P for both parts of the press, P_1 and P_2.
6. Let $(\max(P) = \min(\{P_1, P_2\}))$.

Let $\left(y_1 = \frac{1}{2}l_2\right)$ and $\left(y_2 = l_2 + \frac{1}{2}l_1\right)$.

Also, let $(A_1 = w_1 l_1)$, $(A_2 = w_2 l_2)$, and $(A = A_1 + A_2)$.

Then,

$$\overline{Y} = \frac{A_1 y_1 + A_2 y_2}{A_1 + A_2} \tag{14}$$

Note:

$$I_1 = \frac{1}{12} l_1 w_1^3 + A_1 \left(y_1 - \overline{Y}\right)^2 \tag{15}$$

$$I_1 = \frac{1}{12} l_2 w_2^3 + A_2 \left(y_2 - \overline{Y}\right)^2 \tag{16}$$

Therefore, $(I = I_1 + I_2)$. Let $(M_e := -Pe)$, where $\left(e = l_1 + l_2 + d - \overline{Y}\right)$.

Consider the stress applied to 1 as the compressive stress and the stress applied to 2 as the tensile stress.

Considering the equation for σ_n, we can get the following equivalent equation:

$$\left(\sigma_n = \frac{P_n}{A} - \frac{My}{I} = \frac{P_n}{A} + \frac{P_n e y}{I} = P_n\left(\frac{1}{A} + \frac{ey}{I}\right)\right) \Rightarrow \left(P_n = \sigma_n \left(\frac{1}{A} + \frac{ey}{I}\right)^{-1}\right) \tag{17}$$

Therefore, for section 1, where $\left(y = -\overline{Y}\right)$:

$$P_1 = \sigma_{\text{all, c}} \left(\frac{1}{A} + \frac{-e\overline{Y}}{I} \right)^{-1} = 1.106 * 10^5 \text{ N} \tag{18}$$

Also, for section 2, where $\left(y = l_1 + l_2 - \overline{Y}\right)$

$$P_1 = \sigma_{\text{all, c}} \left(\frac{1}{A} + \frac{-e\left(l_1 + l_2 - \overline{Y}\right)}{I} \right)^{-1} = 9.596 * 10^4 \text{ N} \tag{19}$$

Therefore,

$$\boxed{\max(P) = \min(\{P_1, P_2\}) = 9.596 * 10^4 \text{ N} \tag{20}}$$

4. Appendix Code

The following Jupyter Notebook (attached to the homework submission at "ESC201_HW6.ipynb") was used for calculations.

In [21]:
```
from sympy.solvers import solve
from sympy import Symbol
import numpy as np
```

In [18]:
```
# Problem 4.51

ds = 0.625
d = 4.0
l = 5.5
w = 6.0
Es = 29.0 * 10**6
Ec = 3.0 * 10**6
n = Es / Ec
As = (np.pi / 4)*(ds**2)
sigmac = 1.4 * 10**3
sigmas = 2.0 * 10**4

x = Symbol('x')
y = solve(0.5*l*x**2 - n*As*(d - x), x)
# print(n*As)
print(f"{y[0]:.3e}")
print(f"{y[1]:.3e}")

I = (1 / 3)*l*(y[1]**3) + n*As*(d - y[1])**2
Mc = (I*sigmac) / (y[1])
Ms = (I*sigmas) / (n*(4-y[1]))
print(f"{Mc:.3e}")
print(f"{Ms:.3e}")
```

```
M = (12 / l)*Ms
print(f"{M:.3e}")
```

```
-2.685e+0
1.607e+0
2.143e+4
2.126e+4
4.638e+4
```

In [20]:
```
# Problem 4.65

r = 0.015
D = 0.150
d = 0.100
w = 0.018
sigma = 8 * 10**7
Ka = 1.92
Kb = 1.58
I = (1 / 12)*w*(d**3)

Ma = (2*sigma*I) / (Ka*d)
Mb = (2*sigma*I) / (Kb*d)

print(f"{Ma:.3e}")
print(f"{Mb:.3e}")
```

```
1.250e+03
1.519e+03
```

In [26]:
```
# Problem 4.113

l1 = 0.025
w1 = 0.250
l2 = 0.250
```

```python
w2 = 0.025
d = 0.300
sigmac = -1*8.0 * 10**7
sigmat = 4.0 * 10**7

y1 = 0.5 * l2
y2 = l2 + (0.5 * l1)
A1 = w1*l1
A2 = w2*l2
A = A1 + A2
Y = (A1*y1 + A2*y2) / A

ct = l1 - Y
cc = -1*Y
I1 = (1 / 12)*l1*(w1**3) + A1*(y1 - Y)**2
I2 = (1 / 12)*l2*(w2**3) + A1*(y2 - Y)**2
I = I1 + I2
e = l1 + l2 + d - Y
P1 = sigmac*((A**(-1) + ((e*(-1*Y)) / I))**(-1))
P2 = sigmat*(A**(-1) + ((e*(l1 + l2 - Y)) / I))**(-1)

print(f"{Y:.3e}")
print(f"{I1:.3e}")
print(f"{I2:.3e}")
print(f"{A:.3e}")
print(f"{e:.3e}")
print(f"{P1:.3e}")
print(f"{P2:.3e}")
```

```
1.938e-01
6.209e-05
2.987e-05
1.250e-02
```

```
3.812e-01
1.106e+05
9.596e+04
```